

3.2 Self-Attention (Q, K, V)

Queries ask, keys are asked, values are mixed. Softmax weights over the sequence.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Self-attention is a weighted sum of **values**, with weights from a comparison of **queries** to **keys**. Every token plays all three roles.

1. Q, K, V

For a sequence matrix $X \in \mathbb{R}^{T \times d}$,

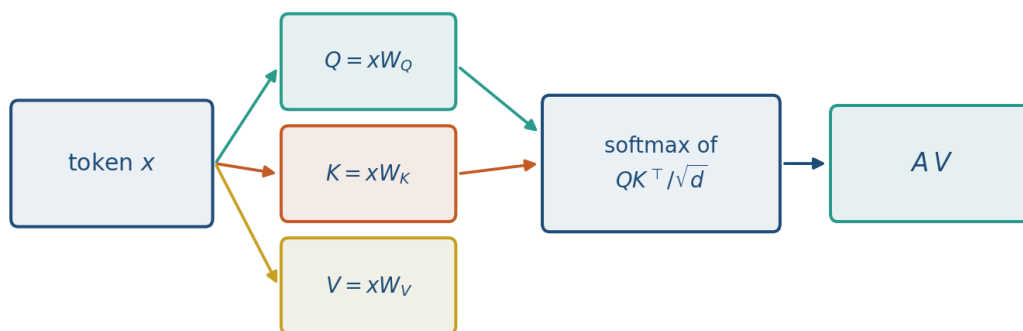
$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Scores are scaled dot products, then a softmax over keys:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \quad \text{Attention}(Q, K, V) = AV.$$

The $\sqrt{d_k}$ keeps the dots from saturating the softmax when d_k is large.

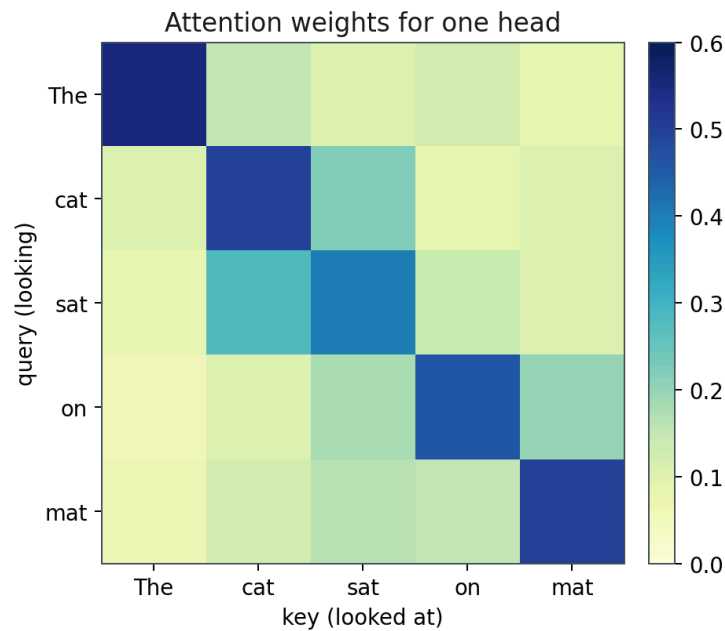
Query asks, key is asked, value is what you mix in



Query: what this token is looking for. **Key:** what another token looks like to be found. **Value:** what you actually add into the representation.

2. A row of weights

Row t of A is a distribution over the sequence. Large mass on a name is "this pronoun just looked at that name."



Multi-head: several (W_Q, W_K, W_V) in parallel, concatenate, project. Different heads can track syntax, a name, or a comma. Lab 3 will compute one head by hand.

3. Masks

For next-token prediction you **mask** future keys (set scores to $-\infty$ before softmax). Otherwise the model cheats. BERT does not mask the future; it masks random tokens and reconstructs them (note 3.4).

4. Practice

1. If two tokens have identical keys, what does the softmax do to their weights?
2. Why divide by $\sqrt{d_k}$?
3. Write A for $T = 2$ with $Q = K = V = I$ (identity), no mask. What is AV ?